Applications of logarithmic functions

1

a 4

b 100

2 $\log \frac{15}{14}$

3 3668 m

4

a $R = \log\left(\frac{x}{a}\right)$

b

i $R = 3.8$

ii Minor

c

i $R = 4.2$

ii Light

5

a 10

b 251

c 16

d 7.3

6

a $1\,000\,000$

b 1000 times louder

7

a

i 127 dB

ii 42 dB

b 19.5

c $L = 80$

d $10 \log 2 \text{ dB}$

e

i 110 dB

ii $I = 10^{-\frac{57}{10}} \text{ watts/cm}^2$

8

a 803 years

b 735 years

c R_1 : 1935 years, R_2 : 1771 years

d No, at twice the half-life there will be one-quarter of the substance left. As the amount of substance approaches zero the time taken for a unit of substance to decay grows at an icreasing rate.

9

a 3.7

b Acidic

c $h = 10^{-8.3}$

10

a $L = 7$

b $N = 62$

c $H = 41.68$ bits

d $N = 16$

11

a $I = \dfrac{1}{10}$ MW

b

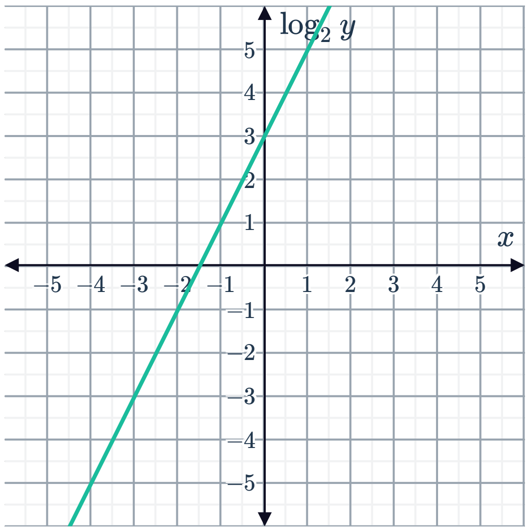
i $(2, 3)$

ii $(6, 7)$

12

a $m = 2, k = 3$

b



c 2

d $x = \dfrac{1}{2}$

Applications of exponential functions

13

a 64

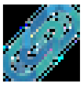
b $12\,288$

c $49\,152$

d The number of reactions is increasing at an increasing rate.

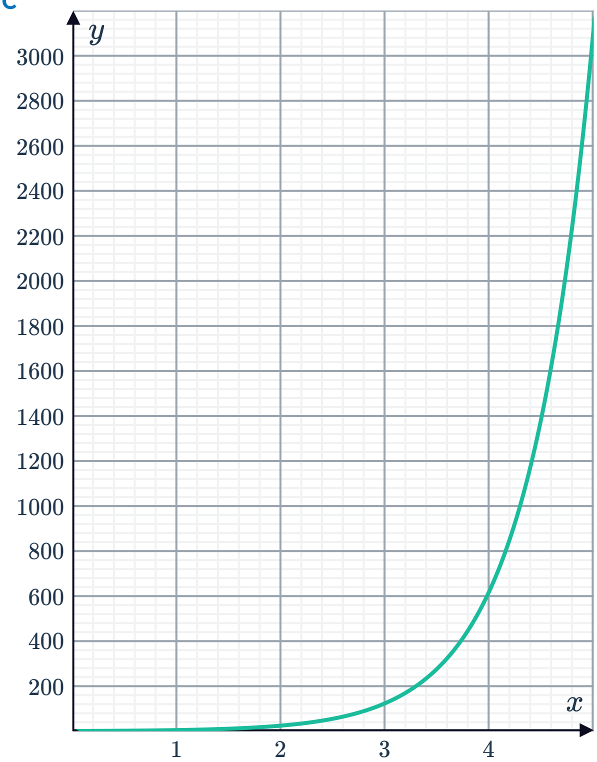
14

a No



b $y = (5)^x$

c



15

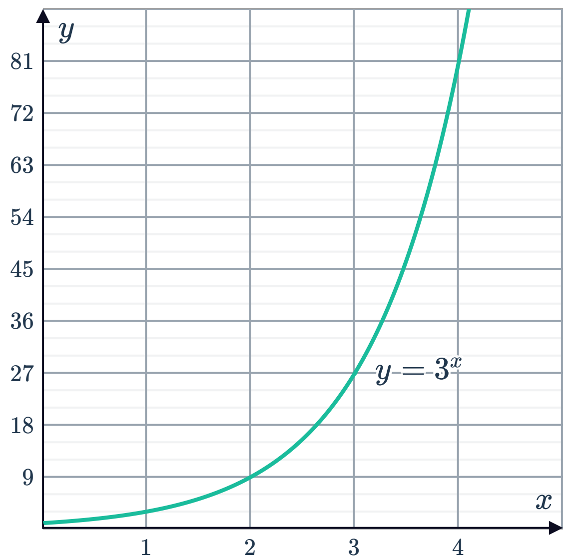
a

Number of hours passed (x)	0	1	2	3	4
Number of bacteria (y)	1	3	9	27	81

b $y = 3^x$

c 177 147

d

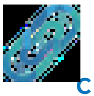


e The initial amount of bacteria, in this case 1.

16

a No

b Yes



c No

17

a

Seconds passed (t)	0	1	2	3	4	5
Number of bacteria (y)	1	2	4	8	16	32

b $y = 2^t$

c 2^{60}

d $y = \left(2^{3600}\right)^h$

18

a

Seconds passed (x)	0	1	2	3	4	5
Undissolved sugar in grams (y)	1	$\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{64}$	$\frac{1}{256}$	$\frac{1}{1024}$

b

i $y = \left(\frac{1}{4}\right)^t$

ii $y = 4^{-t}$

c $\frac{3}{16}$ g

d $\frac{3}{64}$ g

e It is decreasing at a decreasing rate.

f No, the model shows the amount of undissolved sugar is decreasing at a decreasing rate and has an asymptote of $y = 0$.

19

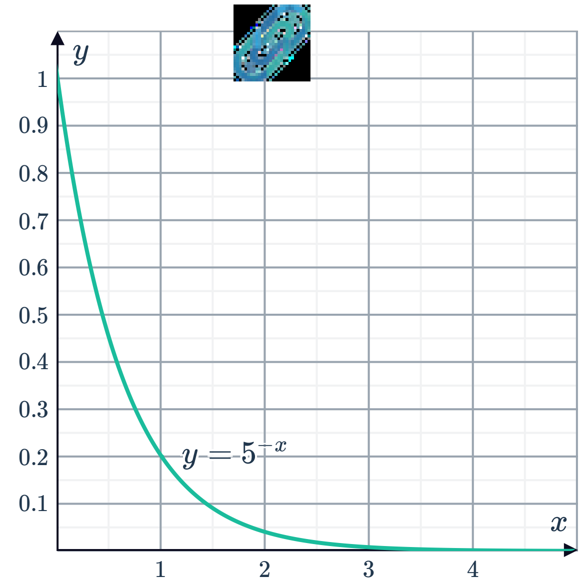
a

Month (x)	0	1	2	3
Amount remaining in kilograms (y)	1	$\frac{1}{5}$	$\frac{1}{25}$	$\frac{1}{125}$

b $y = 5^{-x}$

c $\frac{1}{78\,125}$ kg

d



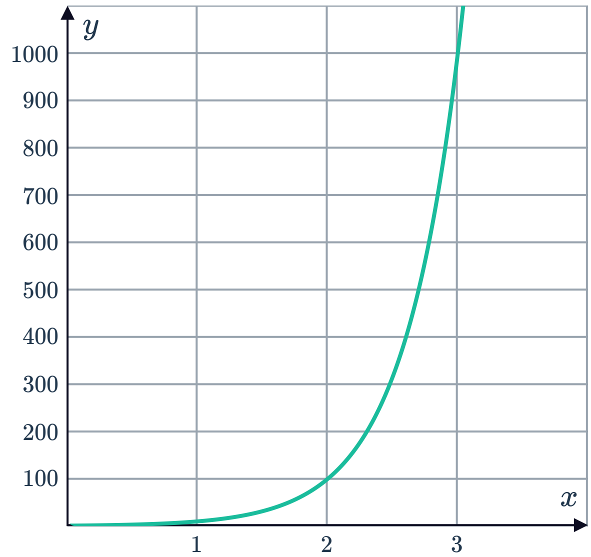
e The initial amount of isotope, in this case 1 kg.

20

a

Richter scale measure (x)	1	2	3	4	5
Microns of ground motion (y)	10	100	1000	10 000	100 000

b

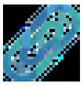


c No, 10^x can never be zero. Also, the function is only defined for $x \geq 0$ with a y-intercept of one and an increasing function.

d 100 times larger

e 1.6 times stronger than aftershock

21



a

Day	0	1	2	3	4
Number of people infected	1	5	25	125	625

b 5^n

c 8 days

22

a $y = (2)^x$

b That there is 1 layer of paper when no folds have been made.

c 1024

d 40.96 mm

23

a 19th day

b $\$1\,597\,152$

24

a

Number of rounds (r)	3	4	5	6
Number of players (p)	8	16	32	64

b Any power of 2.

c $p = 2^r$

d 1024

25

a 8

b 256

c 2^n

26

a 1000 aphids

b $32\,000$ aphids

c 250 aphids

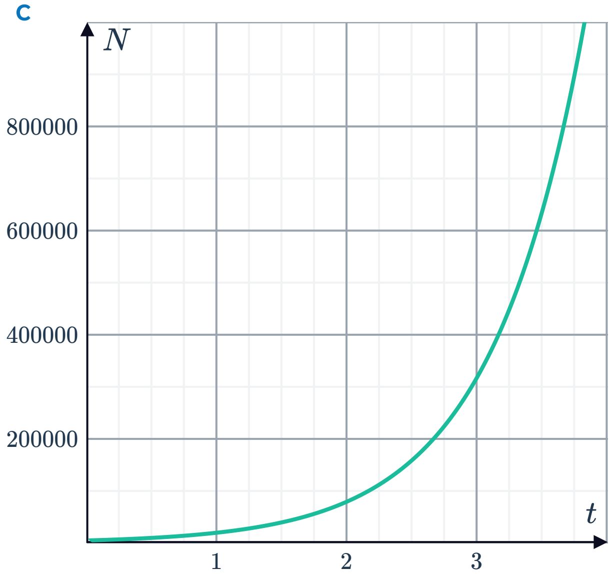
27 $x = 5$

28

a 5000

b 327 680 000

c



d 0.79 hours

29

a 6.7 kg

b 2.0 kg

30

a $35 730.90

b

n	1	2	3	4	5	6
A	32 190.00	35 730.90	39 661.30	44 024.04	48 866.69	54 242.02

c The value of the investment grows at an increasing rate.

31 $241.70

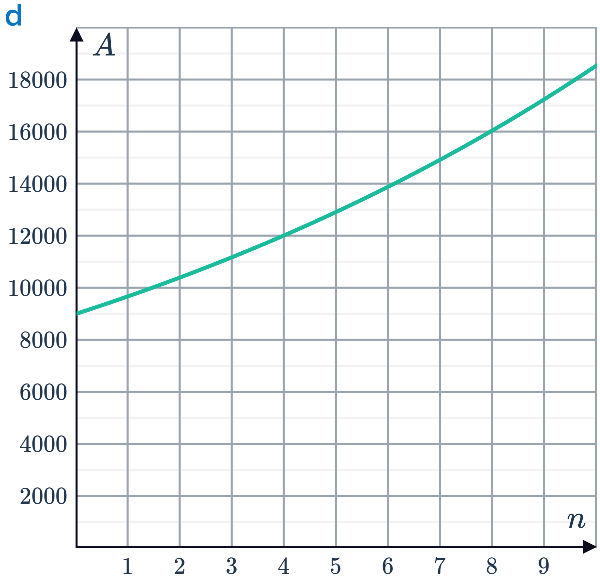
32

a $9000

b 7.5%

c $13 890

d



e 19 years

33

a $V = 2000 \times 1.07^t$

b $2805

34

a

Number of weeks passed (x)	0	1	2	3	4
Number of cells (y)	1728	864	432	216	108

b $a = 2, b = 1728$

c No, the graph has an asymptote of $y = 0$. Although in practice once there is less than one a cell we could consider all cells removed.

35

a Exponential decay at a rate Of 2% per unit of time, with an initial amount of 100.

b Exponential growth at a rate Of 7% per unit of time, with an initial amount of 200.

c Exponential growth at a rate Of 25% per unit of time, with an initial amount of 50.

d Exponential decay at a rate Of 13% per unit of time, with an initial amount of 700.

36

a $2500

b $47 500

c $20 000

d 10

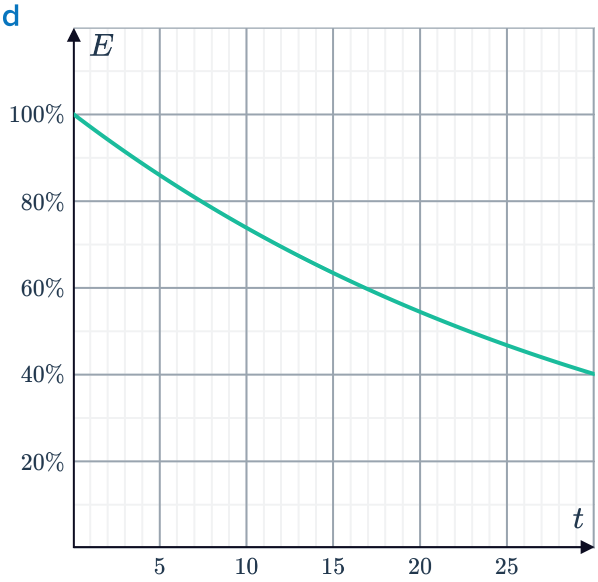
37

a 94.09%

b 76.02%

c $E=(97\%)^{t}$

d



e 23 months

38

a $\$900.90$

b $\$819.82$

c $A=990\,(91\%)^{t}$

d $\$319.25$

39

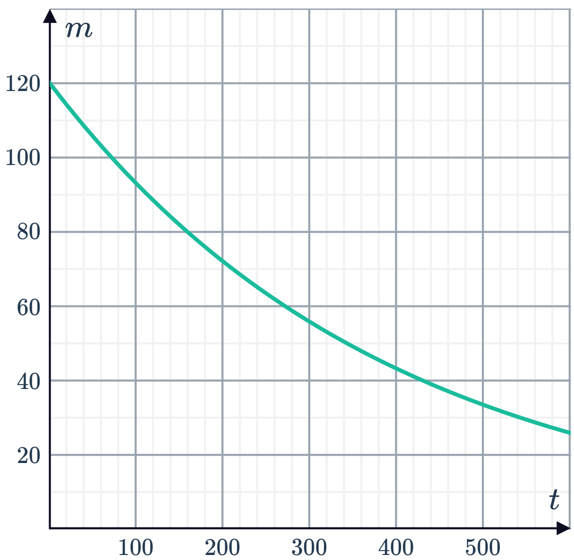
a 105.60 mg

b 92.93 mg

c $m=120\,(88\%)^{\frac{t}{50}}$

d 88.30 mg

e



f 271 days

40

a $k = 57$

b $a = 1.06$

c $P = 57\,(1.06)^n$

d

Year	Population (millions)	Estimated population (millions)	Estimated population - actual population (millions)
1998	57	57.00	0.00
2003	75	76.28	1.28
2008	98	102.08	4.08
2013	127	136.60	9.60
2018	161	182.81	21.81

e 1880.30 million

f No, the prediction is well beyond the range of years the model is based on and we can see from the table that the model appears to be less accurate as time passes.

41

a

Year	1	2	3	4
Number of breeding goats	2	4	8	16
Number of non-breeding goats	9	9	9	9
Total number of goats	11	13	17	25

b 9

c 2

d 16 393

e 10th year